SYNOPISIS

CAPSTONE PROJECT: SMART ENROLL

PG-AIML, CHENNAI
GROUP 2 (2022-23)

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A. Problem Definition

The current process of paper-based health insurance enrollment in the United States is slow, error-prone, and resource-intensive. This outdated process not only increases the administrative burden for health insurance providers but also causes frustration for customers, resulting in longer wait times and reduced satisfaction levels. With over 330 million people in the USA, the scale of this problem is enormous, with millions of paper-based applications being processed each year.

Opportunity

To address this issue, we propose to digitize the health insurance enrollment process. By automating this process, we can reduce the need for manual data entry, which can minimize errors and improve processing times. According to the National Association of Insurance Commissioners, in 2020, there were approximately 242 million people with private health insurance and another 74 million with Medicaid coverage in the United States.

With an estimated 316 million people insured in the USA, the potential financial opportunity for this business case is significant. By reducing the administrative burden associated with paper-based enrollment, health insurance providers can streamline their operations, reduce costs, and improve customer satisfaction levels. Additionally, by leveraging AI tools to digitize this process, insurers can ensure that they are complying with regulatory requirements and keeping up with industry trends towards digital transformation.

Solution

The opportunity to digitize paper-based health insurance enrollment using AI tools is significant, with the potential to deliver tangible benefits for both health insurance providers and their customers. Our **SMART Enroll project** would involve using state-of-the-art transformer-based AI models to automatically extract and process information from paper-based applications. By leveraging the power of these models, we can minimize errors, improve processing times, and create a seamless and efficient enrollment process for health insurance providers and customers alike. Our SMART Enroll Project represents a significant step towards digital transformation for the USA healthcare industry to begin with but extensible to a wide variety of use-cases that involve Form-based enrollment.

The 4 (four) components of this Project are shown in Figure 1.

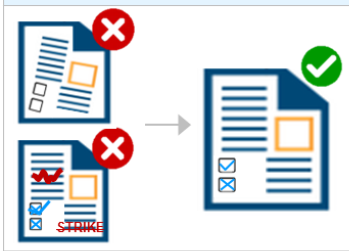
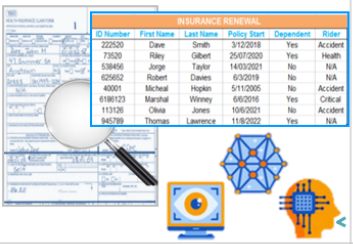
SCAN PRE-PROCESSING	PRE-VALIDATION & SORTING	AIML MODEL: CV + NLP	VALIDATION & ANALYTICS
			
▪ Tilted / Inverted scan / Scaling ▪ Color / Black & White scans ▪ Handle Form Field Segmentation ▪ Noise Reduction & Cleaning ▪ Single / Batch File	▪ Flag Anomalies [wrong data, Age Rule 18 ~ 65, 10 Digit Phone, etc] ▪ Sorting: Insurance - Proposal type [Health - New / Health - Claims, etc]	▪ OCR: Text Detection ▪ ICR: use pre-trained Model for Handwriting [Characters / Digits] ▪ Relevant Data Extraction ▪ Structured Data Format	▪ Business Critical Data Validation ▪ Analyze Downstream [Classify: Claim Approve / Reject]

Figure 1: Major components to SMART Enroll the Health Insurance Forms

MVP Scope is planned for the Health Insurance Forms of USA based major Health Insurer, Highmark Health Inc.

Future Scope: To scale for various other Insurance Company Form Templates and other proposal types like Claim / / Renewal / Annuity. This could also include multi-lingual data.

B. Literature Survey

Data Extraction Challenges

- Orientation / Skew
- Check-box identification
- Varying Form Layouts and Context
- Handling Handwritten Text with multiple Styles, Fonts and appearances

Identify the Form Layout structure and distinguish between labels and key value pair versus the additional features / Noise like Paragraphs, Images, Logos, etc that is not required for the structured data output format.

The complex problem of Handwriting Recognition:

- Huge variability and ambiguity of strokes from person to person
- Handwriting style of an individual person also varies time to time and is inconsistent
- Poor quality of the source document/image due to degradation over time
- Text in handwriting can have variable rotation to the right contrast to the Straight printed text
- Cursive handwriting makes separation and recognition of characters challenging
- Collecting a good labeled dataset to learn is not cheap compared to synthetic data

Study of Comp. Vision / Handwritten Text Recognition / Information Extraction Models

Choice of the AIML Model should be based on specific task and the available Data. However, considering our Project Future scope and Scaling of the Model, especially understanding the Document Layout, we have considered the most recent advancements in the Computer Vision (CV) and Natural Language Processing (NLP) domain. Transformer-based models have shown to be effective in recognizing handwritten text, due to their attention mechanism, transfer learning capabilities, and availability of pre-trained models.

- **Attention Mechanism:** The self-attention mechanism in transformers allows the model to focus on relevant parts of the input sequence, making it effective in recognizing patterns in handwritten text. This mechanism also allows the model to handle inputs of different lengths and to capture long-range dependencies.

- **Transfer Learning:** Pre-trained transformer models, such as BERT and GPT, can be fine-tuned for a wide range of tasks, including recognizing handwritten text, with minimal additional training. This capability allows the models to leverage knowledge learned from large amounts of data, which can improve their performance on the target task.

Plan of Experiments

We plan to use transformer-based models for information extraction in a structured format. These models pre-trained on large amounts of text data, are to be fine-tuned on the labeled data to extract the relevant fields and their values. The extracted data can then be populated in a structured database, which can be used for further analysis and decision-making.

To evaluate the suitability for the SMART enrollment of the Forms, we plan to experiment with the following Transformer architecture methods:

1. **Use Optical Character Recognition (OCR) API for Annotations [Text detection and Text Recognition].** We plan to label the fields with the corresponding data types, such as text, number, date, checkbox, using a labeling tool. The labeling / Annotation process is planned to be used as ground truth for AI algorithm performance evaluation.
2. However, only OCR method can lead to Low accuracy with Text recognition errors due to Hand-written Text with variations in Font styles, sizes, appearances and also their layouts. This requires Natural Language Processing (NLP) for Handwritten Text Recognition (HTR) with High accuracy.
3. **Experiment with off-the-shelf OCR APIs and use the OCR API Output as input to the Transformer Model for further fine-tuning:**
    - Transfer Learn: Initialize StructuralLM Model weights using the Benchmark FUNSD Dataset
    - Convert the Text into Token Embeddings and use relative 2D Positional Embeddings
    - Training & Fine-tuning**:** Training on Health Insurance Forms (HIFI) Dataset; Generate Tokens and use Ground Truth [Annotations / BBoxes]
4. **Experiment with Multimodal Transformer:**
    - Transfer Learn: Initialize LayoutLMv3 Model weights using the Benchmark FUNSD Dataset
    - Use OCR for Word, 1D & 2D Position Embeddings and Image Embeddings.
    - Training & Fine-tuning**:** Training on Health Insurance Forms (HIFI) Dataset
5. **Experiment with OCR-free Transformer Encoder to extract features from the Document and Transformer Textual Decoder to map features with Tokens**
    - Transfer Learn: Initialize DONUT Model weights using the Benchmark IIT-CDIP Dataset
    - Use Image Embeddings from Swin Transformer / ViT / Faster R-CNN
    - Generate Token Sequences from Decoder

- Training & Fine-tuning: Training on Insurance Forms (HIFI) Dataset

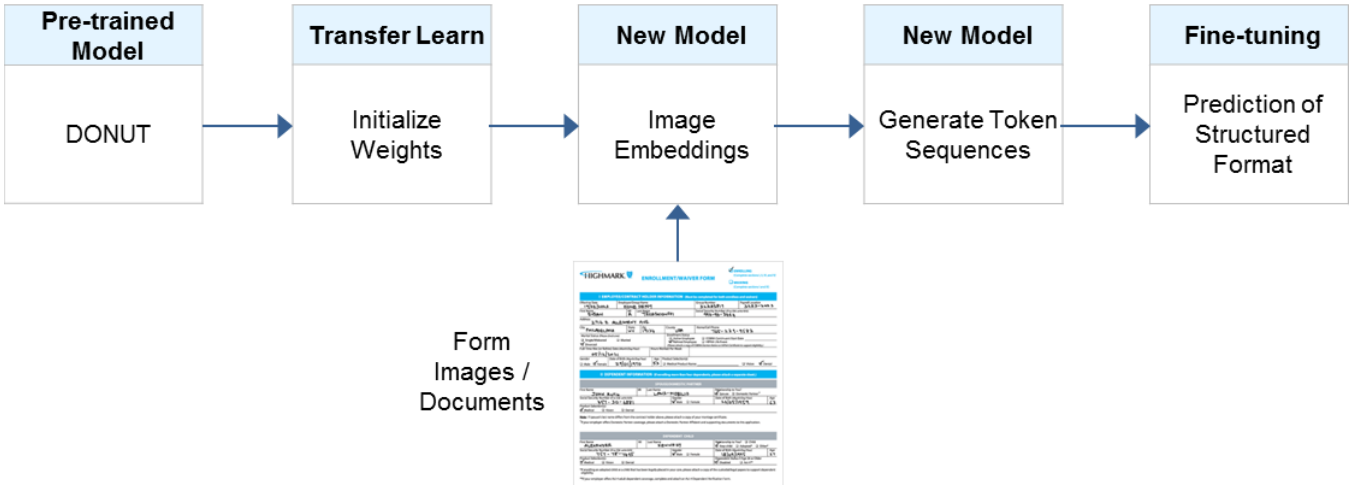


Figure 2: Experiment Pipeline with OCR-free Transformer Model

6. Utilize the above trained Models for Information Extraction (IE) application from our HIFI Dataset
7. **Testing:** Our trained Models [Smart Enroll #1, #2, #3] vs. BERT, BROS, StructuralLM, LayoutLMv3, DONUT on our Health Insurance Form Info (HIFI) Dataset IE task
8. **Model Performance Evaluation:** Key metrics used for IE applications are f1-score and TED accuracy (Tree Edit Distance)

Table 1: Sample Plan - IE Experiment Evaluation on Health Insurance Forms Information (HIFI)

		Health Insurance Forms Info (HIFI) Dataset)		
Model	**Parameters (M)**	**f1-Score**	**TED Accuracy**	**Time**
BERT				
BROS				
LayoutLMv3				
DONUT				
SMART Enroll #1				
SMART Enroll #2				
SMART Enroll #3				

C. Dataset

Data Description and Preparation Method

Demographics of Insurance Product subscribers and their Families collected in the Health Insurance Form of *Highmark Health Inc*, USA's largest Blue Cross and Blue Shield Insurer.

The insurance enquiry **Health Insurance Form Information** (**HIFI**) dataset used in this study consists of 300 one-page forms. The dataset contains personal information required for insurance enrollment, including spouse and child details. There are a total of 40 fields per form. Sample of scanned Forms can be seen in Figure 3.

HIFI Dataset is made up of 300 impersonated individual information. The dataset has been curated to contain heterogeneity in the data to resemble real world scenarios:

- Some Special Name styles with Hyphens / Accent Marks
- Varied demographics combinations e.g. Gender, Marital status, Employment status, Age, etc
- Social Security Number Format 111-22-3333
- Date Format MM/DD/YYYY
- Phone Number Format 111-222-3333.
- All the Handwritten Text is filled in Capital case.
- Checkbox Data (Tick-marks) for Gender, Product Type, etc

In total, 300 Forms [70% Color and 30% Black & White] were filled by various Persons and therefore a corpus of 50+ Handwriting Styles is in this Dataset. To create the Handwritten Forms, High Resolution Scans (300 to 600 dpi) and Data Verification took a consolidated fortnight's time [10 to 25 Apr 2023].

Google Drive Link to the Data:
https://drive.google.com/drive/folders/1rO3N22qnxrzEI3SZJ1OwwTniReLLCIVJ?usp=sharing

Metadata

Personal Details: Name, Social Security No., Address [City, State, Zip], Phone No., Marital Status, Gender, DOB, Age

EFF_DATE	SUB_FIRST_NAME	MID_IN	SUB_LAST_NAME	SUB_SSN	SUB_ADDRESS	SUB_CITY	SUB_STATE	SUB_ZIP	SUB_PHONE	SUB_MARITAL STATUS	SUB_GENDER	SUB_DOB	SUB_AGE
05/01/2023	COLBY	A	EISENBACH	757-78-5749	2115 DUNCAN REGIONAL LOOP	DUNCAN	AL	73533	765-239-9487	SINGLE/WIDOWED	MALE	01/01/1987	36
05/01/2023	BARRY	C	WATERS	190-90-3464	9750 NW 33RD ST	CORAL SPRINGS	AZ	33065	765-239-9489	DIVORCED	MALE	03/01/1955	68
05/01/2023	TIMOTHY	X	O'KIEFE	190-90-3464	9947 MOLLY PITCHER HWY	SHIPPENSBURG	MI	17257	765-239-9510	DIVORCED	MALE	05/27/1954	69
05/01/2023	BABAK	A	ATTARAN-REZAEI	980-51-4189	13737 NOEL RD	DALLAS	NY	75240	765-239-9567	SINGLE/WIDOWED	MALE	01/01/1975	48
05/01/2023	DUMITRU-OCTAVIAN		SANDULESCU	775-78-5749	38300 VAN DYKE AVE	STERLING HEIGHTS	NE	48312	765-239-9660	DIVORCED	MALE	05/19/1951	72
06/01/2023	FÉLICITÉ		HANDSHOE	757-78-5749	4915 NORTON HEALTHCARE BLVD	LOUISVILLE	UT	40241	765-239-9677	MARRIED	FEMALE	09/19/1987	36
19/04/2023	SUSAN	A	TRZESNIOWSKI	990-90-3464	2716 E ALLEGHENY AVE	PHILADELPHIA	WV	19134	765-239-9583	DIVORCED	FEMALE	29/01/1970	53
01/03/2023	MICHAEL	F	PATNAUDE	214-91-3731	50 PARK RD	WESTBROOK	NV	4092	765-239-9612	MARRIED	MALE	27/02/1987	36

Employment Details: Employer, Payroll Location, Employment Type, Enrollment Status, Hire Date (MM/DD/YYYY), Insurance Product Type

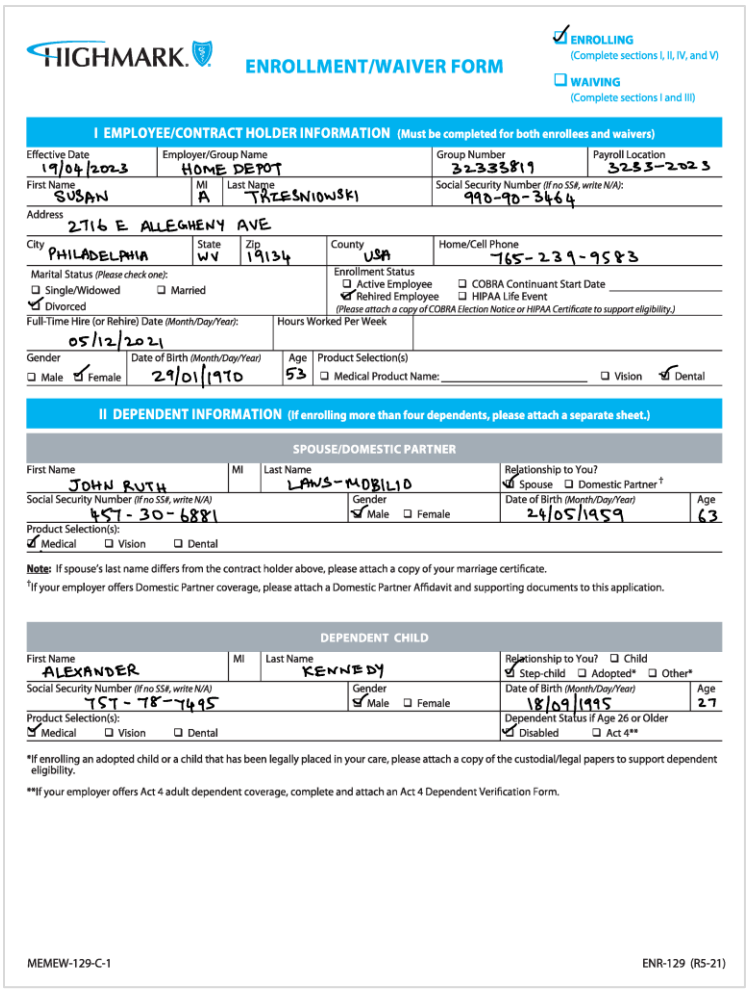
SUB_FIRST_NAME	GRP_NAME	GRP_NO.	PAYROLL_LOCATION	ENROLLMENT STATUS	HIRE_DATE	HRS_PER_WEEK
COLBY	UNITED HEALTH GROUP	01245729	0124-2023	ACTIVE EMPLOYEE	06/16/2000	40
BARRY	STARBUCKS	47289191		ACTIVE EMPLOYEE	03/05/2020	30
TIMOTHY	UNITED HEALTH GROUP	01245729	0124-2023	ACTIVE EMPLOYEE	03/08/2011	45
BABAK	STARBUCKS	47289191		REHIRED EMPLOYEE	05/19/2022	36
DUMITRU-OCTAVIAN	HOME DEPOT	32333819	3233-2023	ACTIVE EMPLOYEE	08/20/2000	
FÉLICITÉ	STARBUCKS	47289191		ACTIVE EMPLOYEE	03/15/2020	60
SUSAN	HOME DEPOT	32333819	3233-2023	REHIRED EMPLOYEE	05/12/2021	
MICHAEL	UNITED PARCEL SERVICE	17381991		COBRA CONTINUANT START DATE	11/11/2022	

Dependent Details: Relationship with Subscriber, Spouse & Child Details, Dependent Status, Product

SUB_FIRST_NAME	SPO_ FIRST_NAME	SPO_ MID_INIT	SPO_ LAST_NAME	RELN_ TO_SUBR	SPO_ SSN	SPO_ GENDER	SPO_ DOB	SPO_ AGE	SPO_ PRODUCT
COLBY									
BARRY	PAOLA	N	WATERS	DOMESTIC PARTNER	980-51-4189	FEMALE	04/27/1979	44	DENTAL
TIMOTHY	LAURA		O'KIEFE	SPOUSE	757-78-5749	FEMALE			DENTAL
BABAK									DENTAL
DUMITRU-OCTAVIAN									
FÉLICITÉ									
SUSAN	JOHN RUTH		LAWS-MOBILIO	SPOUSE	457-30-6881	MALE	24/05/1959	63	MEDICAL
MICHAEL	BRENNA		RADERMACHER	SPOUSE	775-78-5749	FEMALE	17/10/1991	31	MEDICAL

SUB_FIRST_NAME	CHILD_ FIRST_NAME	CHILD_ MID_INIT	CHILD_ LAST_NAME	CHILD_ RELN_TO_SU	CHILD_ SSN	CHILD_ GENDER	CHILD_ DOB	CHILD_ AGE	CHILD_ PRODUCT	CHILD_ DEPN_STATUS
COLBY										
BARRY	RICHARD	S	WATERS	STEP-CHILD	757-78-5749	MALE	01/01/2000	23	DENTAL	
TIMOTHY	TAMAR		O'KIEFE	CHILD	114-91-3731	FEMALE	01/01/1998	25	DENTAL	
BABAK										
DUMITRU-OCTAVIAN										
FÉLICITÉ										
SUSAN	ALEXANDER		KENNEDY	STEP-CHILD	757-78-7495	MALE	18/09/1995	27	MEDICAL	DISABLED
MICHAEL										

Figure 3: Samples of Scanned Handwritten Form(s)



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ENROLLMENT/WAIVER FORM

☑ ENROLLING (Complete sections I, II, IV, and V)

☐ WAIVING (Complete sections I and III)

I EMPLOYEE/CONTRACT HOLDER INFORMATION (Must be completed for both enrollees and waivers)

Effective Date	Employer/Group Name	Group Number	Payroll Location
19/04/2023	HOME DEPOT	32333819	3233-2023

First Name	MI	Last Name	Social Security Number (If no SS#, write N/A):
SUSAN	A	TRZESNIOWSKI	990-90-3464

Address: 2716 E ALLEGHENY AVE

City	State	Zip	County	Home/Cell Phone
PHILADELPHIA	WV	19134	USA	765-239-9583

Marital Status (Please check one): ☐ Single/Widowed ☐ Married ☑ Divorced

Enrollment Status: ☐ Active Employee ☐ COBRA Continuant Start Date ☑ Rehired Employee ☐ HIPAA Life Event (Please attach a copy of COBRA Election Notice or HIPAA Certificate to support eligibility.)

Full-Time Hire (or Rehire) Date (Month/Day/Year):	Hours Worked Per Week
05/12/2021	

Gender	Date of Birth (Month/Day/Year)	Age	Product Selection(s)
☐ Male ☑ Female	29/01/1970	53	☐ Medical Product Name: _____ ☐ Vision ☑ Dental

II DEPENDENT INFORMATION (If enrolling more than four dependents, please attach a separate sheet.)

SPOUSE/DOMESTIC PARTNER

First Name	MI	Last Name	Relationship to You?
JOHN RUTH		LAWS-MOBILD	☑ Spouse ☐ Domestic Partner†

Social Security Number (If no SS#, write N/A)	Gender	Date of Birth (Month/Day/Year)	Age
457-30-6881	☑ Male ☐ Female	24/05/1959	63

Product Selection(s): ☑ Medical ☐ Vision ☐ Dental

Note: If spouse's last name differs from the contract holder above, please attach a copy of your marriage certificate.

†If your employer offers Domestic Partner coverage, please attach a Domestic Partner Affidavit and supporting documents to this application.

DEPENDENT CHILD

First Name	MI	Last Name	Relationship to You? ☐ Child
ALEXANDER		KENNEDY	☑ Step-child ☐ Adopted* ☐ Other*

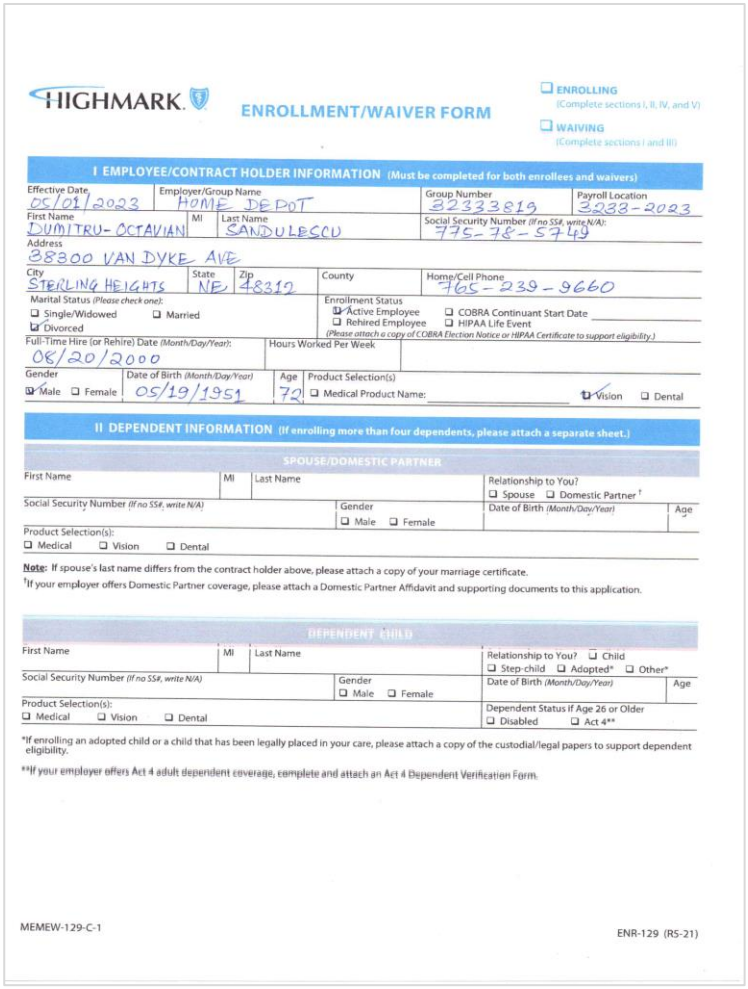
Social Security Number (If no SS#, write N/A)	Gender	Date of Birth (Month/Day/Year)	Age
757-78-7495	☑ Male ☐ Female	18/09/1995	27

Product Selection(s):	Dependent Status if Age 26 or Older
☑ Medical ☐ Vision ☐ Dental	☑ Disabled ☐ Act 4**

*If enrolling an adopted child or a child that has been legally placed in your care, please attach a copy of the custodial/legal papers to support dependent eligibility.

**If your employer offers Act 4 adult dependent coverage, complete and attach an Act 4 Dependent Verification Form.

MEMEW-129-C-1 ENR-129 (RS-21)



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ENROLLMENT/WAIVER FORM

☐ ENROLLING (Complete sections I, II, IV, and V)

☐ WAIVING (Complete sections I and III)

I EMPLOYEE/CONTRACT HOLDER INFORMATION (Must be completed for both enrollees and waivers)

Effective Date	Employer/Group Name	Group Number	Payroll Location
05/01/2023	HOME DEPOT	32333819	3233-2023

First Name	MI	Last Name	Social Security Number (If no SS#, write N/A):
DUMITRU-OCTAVIAN		SANDULESCU	775-78-5749

Address: 38300 VAN DYKE AVE

City	State	Zip	County	Home/Cell Phone
STERLING HEIGHTS	MI	48312		765-239-9660

Marital Status (Please check one): ☐ Single/Widowed ☐ Married ☑ Divorced

Enrollment Status: ☑ Active Employee ☐ COBRA Continuant Start Date ☐ Rehired Employee ☐ HIPAA Life Event (Please attach a copy of COBRA Election Notice or HIPAA Certificate to support eligibility.)

Full-Time Hire (or Rehire) Date (Month/Day/Year):	Hours Worked Per Week
08/20/2000	

Gender	Date of Birth (Month/Day/Year)	Age	Product Selection(s)
☑ Male ☐ Female	05/19/1951	72	☐ Medical Product Name: _____ ☑ Vision ☐ Dental

II DEPENDENT INFORMATION (If enrolling more than four dependents, please attach a separate sheet.)

SPOUSE/DOMESTIC PARTNER

First Name	MI	Last Name	Relationship to You?
			☐ Spouse ☐ Domestic Partner†

Social Security Number (If no SS#, write N/A)	Gender	Date of Birth (Month/Day/Year)	Age
	☐ Male ☐ Female		

Product Selection(s): ☐ Medical ☐ Vision ☐ Dental

Note: If spouse's last name differs from the contract holder above, please attach a copy of your marriage certificate.

†If your employer offers Domestic Partner coverage, please attach a Domestic Partner Affidavit and supporting documents to this application.

DEPENDENT CHILD

First Name	MI	Last Name	Relationship to You? ☐ Child
			☐ Step-child ☐ Adopted* ☐ Other*

Social Security Number (If no SS#, write N/A)	Gender	Date of Birth (Month/Day/Year)	Age
	☐ Male ☐ Female		

Product Selection(s):	Dependent Status if Age 26 or Older
☐ Medical ☐ Vision ☐ Dental	☐ Disabled ☐ Act 4**

*If enrolling an adopted child or a child that has been legally placed in your care, please attach a copy of the custodial/legal papers to support dependent eligibility.

**If your employer offers Act 4 adult dependent coverage, complete and attach an Act 4 Dependent Verification Form.

MEMEW-129-C-1 ENR-129 (RS-21)

D. List of Algorithms

We intend to evaluate our HIFI dataset against the benchmark and latest Transformer Models using the Key metrics used for IE applications like f1-score and TED accuracy.

- Structural LM
- Transformer Multimodal Model: LayoutLMv3
- Swin Transformer / Vision Transformer / Faster R-CNN
- BRAT Model
- OCR free Document Understanding Transformer [DONUT] Model
- BROS Model

		Health Insurance Forms Info (HIFI) Dataset)		
Model	**Parameters (M)**	**f1-Score**	**TED Accuracy**	**Time**
BERT				
BROS				
LayoutLMv3				
DONUT				
SMART Enroll #1				
SMART Enroll #2				
SMART Enroll #3				

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